

Atlas

+

Voyage AI

+

Vector Search

HEALTHCARE AI DEMO

AI-assisted prior *authorization*.

DOMAIN

Payer · PA adjudication

STACK

Atlas · Voyage AI · Vector Search

FORMAT

~10 min product tour

THE PROBLEM

Pended claims sit in a queue while reviewers hunt through policy PDFs, prior cases, and unstructured clinical notes to make a medical-necessity call.

The data needed to adjudicate already exists — coverage policies, comparable approved/denied cases, and the member's own clinical summary. The work is finding it and assembling it into a defensible recommendation.

~3

business days, typical PA turnaround — mostly review queue and document retrieval

100s

of coverage policies and prior cases per line of business, growing every quarter

1

document the reviewer ultimately updates — the claim record itself

THE STACK

Three components, one document.

01

Atlas

OPERATIONAL DATA LAYER

Holds claims, coverage policies, prior cases, and the AI rationale — in the same flexible document that the adjudication system already reads and writes.

- Flexible document schema
- Vectors stored on the record
- Same DB as the app

02

Voyage AI

EMBEDDING MODEL

voyage-3 turns the clinical notes attached to a pended claim into a 1024-dim vector that captures clinical meaning — not just keywords.

- voyage-3 · 1024 dims
- Clinical-domain quality
- Embeds on demand

03

Vector Search

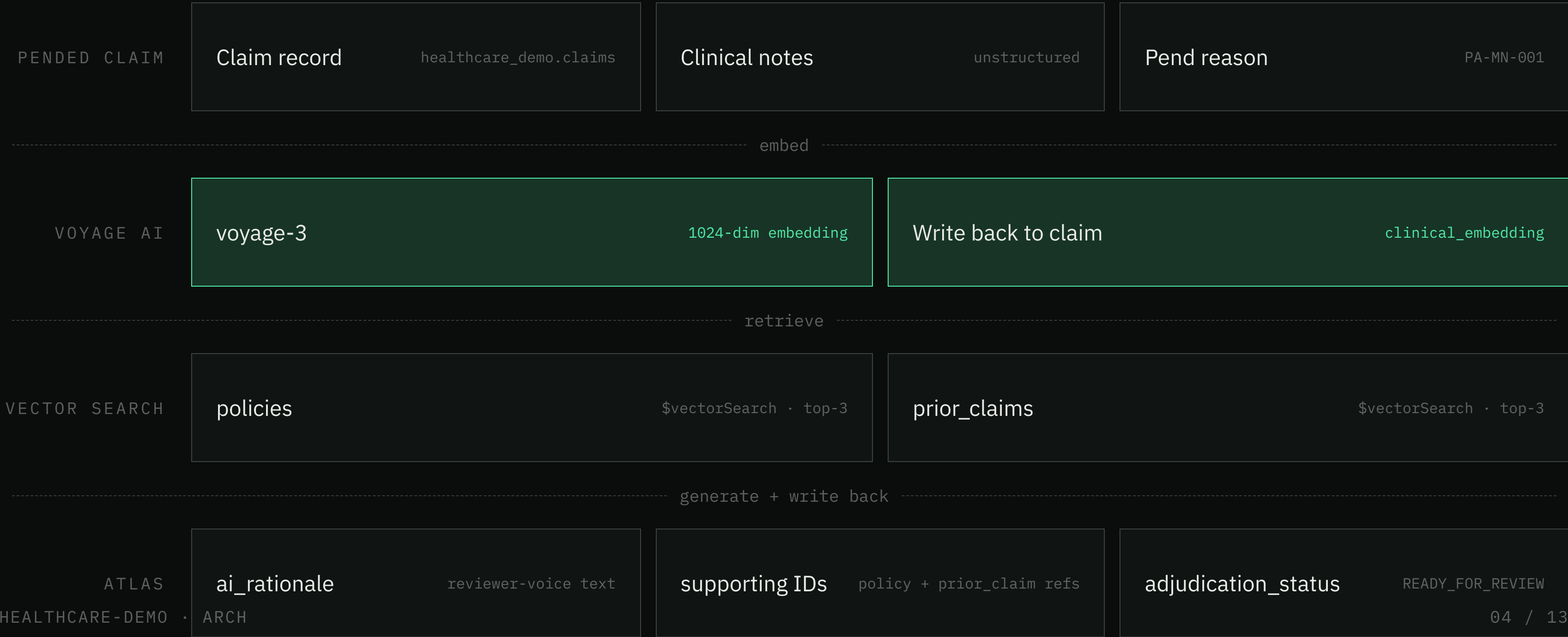
RETRIEVAL ENGINE

One `$vectorSearch` stage finds the most relevant policies and comparable prior cases — with hard filters on plan, state, and outcome.

- Semantic + hard filters
- Native aggregation pipeline
- No separate vector store

ARCHITECTURE

From pended claim to ready for review.



PART · 02

The *walkthrough*.

Scenario picker → operational record → embedding → vector search → retrieved context → rationale & write-back.

WALKTHROUGH

SCREEN 01 / 07 Three pending claims.

MongoDB Atlas

Healthcare AI Demo · Voyage AI + Vector Search

One platform · Operational + AI

Docs ▾

Reset Demo

SELECT DEMO SCENARIO

SCENARIO A · Pend: PA-MN-001

MRI Prior Authorization

Lumbar Spine · CPT 72148

58-year-old male, PPO/OH. Pending for medical-necessity review. 14-week lumbar radiculopathy with conservative treatment history.

SCENARIO B · Pend: MN-DRUG-HCB-002

Specialty Drug Infusion

Infliximab (Remicade) · HCPCS J1745

44-year-old female, PPO/TX. High-cost biologic (\$18,420). Step-therapy compliance review triggered by adjudication engine.

SCENARIO C · Pend: PA-OBE-GLP1-001

GLP-1 Prior Authorization

Semaglutide (Wegovy) 2.4mg · HCPCS S0148

52-year-old female, PPO/FL. Morbid obesity (BMI 38.2) with T2DM, hypertension, and sleep apnea. Pending for lifestyle program documentation.

↑

Select a demo scenario above to begin.

Scenario A — MRI Prior Auth · Scenario B — Infliximab Infusion · Scenario C — GLP-1 / Semaglutide

WALKTHROUGH

SCREEN 02 / 07

One document, one record.

Member, plan, diagnosis, procedure codes, adjudication status, and the unstructured clinical review note — all on the same claim document. This is what the AI pipeline reads from, and what it will write back to.

STRUCTURED Member, plan, ICD-10, CPT, billed amount, pend reason

UNSTRUCTURED UM clinical review note attached to the claim

STATUS adjudication_status: PENDED · pend_reason: PA-MN-001

- Atlas
- flexible schema
- single document

demo.app / claim / record

MongoDB Atlas Healthcare AI Demo · Voyage AI + Vector Search

One platform · Operational + AI Docs Reset Demo

SELECT DEMO SCENARIO

SCENARIO A · Pend: PA-MN-001 ACTIVE

MRI Prior Authorization

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SCENARIO B · Pend: MN-DRUG-HCB-002

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CLM-2025-0001847 James R. Thornton PPO · OH

PEND_REASON PA-MN-001

CLINICAL_EMBEDDING stored · 1024 dims

ADJUDICATION_STATUS PENDED

1 Operational Record 2 Voyage Embedding 3 Vector Search 4 Retrieved Context 5 Rationale + Write-back

Claim Record healthcare_demo.claims PENDED CLM-2025-0001847

MEMBER & PLAN

member_name James R. Thornton

member_id HUM-OH-4482019-01

plan_name Humana Commercial PPO Gold

plan_type / state PPO · OH

DIAGNOSIS

primary_diagnosis_code M51.16

primary_diagnosis_description Intervertebral disc degeneration, lumbar region

PROCEDURE CODES

CPT 72148 MRI Lumbar Spine without contrast · \$4,250

total_billed_amount \$4,250.00

ADJUDICATION

adjudication_status PENDED

pend_reason_code PA-MN-001

pend_reason_description Prior Authorization Required — Medical Necessity Not Established. Physical therapy documentation submitted is incomplete (missing initial evaluation and functional outcome measures). Additional clinical records required per plan policy POL-MRI-LUMBAR-2024-03.

VOYAGE AI EMBEDDING (STORED IN THIS DOCUMENT)

clinical_embedding <vector:1024 dims> · model: voyage-3

embedding_generated_at 2026-05-21T22:08:03.625910+00:00

CLINICAL NOTES (UNSTRUCTURED — EMBEDDED BY VOYAGE AI)

UTILIZATION MANAGEMENT CLINICAL REVIEW NOTE

Date: May 16, 2025 | Reviewer: Patricia Chen, RN, CCM

Member: James R. Thornton (HUM-OH-4482019-01)

Auth Request: Lumbar Spine MRI without contrast (CPT 72148)

CLINICAL SUMMARY:

Member is a 58-year-old male with a 14-week history of progressive low back pain with left lower-extremity radiation in an L4-L5 dermatomal distribution, consistent with lumbar disc herniation causing radiculopathy. Onset followed a workplace lifting incident in February 2025. Ordering physician Dr. Michael Saenz (Internal Medicine) documents worsening functional limitation despite a structured multimodal conservative treatment course.

FIG. 02 · OPERATIONAL RECORD

SCREENSHOT

HEALTHCARE-DEMO · RECORD

07 / 13

WALKTHROUGH

 **MongoDB Atlas**

Healthcare AI Demo · Voyage AI + Vector Search

One platform · Operational + AI

Docs ▾

Reset Demo

SELECT DEMO SCENARIO

SCENARIO A · Pend: PA-MN-001

ACTIVE

MRI Prior Authorization

Lumbar Spine · CPT 72148

58-year-old male, PPO/OH. Pended for medical-necessity review. 14-week lumbar radiculopathy with conservative treatment history.

SCENARIO B · Pend: MN-DRUG-HCB-002

Specialty Drug Infusion

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SCENARIO C · Pend: PA-OBE-GLP1-001

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CLM-2025-0001847

James R. Thornton

PPO · OH

PEND_REASON

PA-MN-001

CLINICAL_EMBEDDING

stored · 1024 dims

ADJUDICATION_STATUS

PENDED

1 Operational Record

2 Voyage Embedding

3 Vector Search

4 Retrieved Context

5 Rationale + Write-back

2 Voyage AI Embedding

voyage-3 · 1024 dims · stored in MongoDB

Generate Embedding

Click **Generate Embedding** to have Voyage AI embed the clinical notes and write the vector directly into this MongoDB document — no separate vector store.

HEALTHCARE-DEMO · EMBED

08 / 13

WALKTHROUGH

Semantic similarity + hard filters.

Vector Search

\$vectorSearch

MongoDB Atlas

Healthcare AI Demo · Voyage AI + Vector Search

One platform · Operational + AI

Docs ▾

Reset Demo

SELECT DEMO SCENARIO

SCENARIO A · Pend: PA-MN-001

ACTIVE

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CLM-2025-0001847

James R. Thornton

PPO · OH

PEND_REASON

PA-MN-001

CLINICAL_EMBEDDING

stored · 1024 dims

ADJUDICATION_STATUS

PENDED

1 Operational Record

✓ Voyage Embedding

3 Vector Search

4 Retrieved Context

5 Rationale + Write-back

3 Atlas Vector Search

Semantic similarity + hard metadata filters

Atlas Vector Search finds semantically relevant policies and prior cases while honoring the filters below as hard constraints — not hints.

PLAN_TYPE

PPO ▾

STATE

OH ▾

ADJUDICATION_OUTCOME (PRIOR CLAIMS)

Any (approved + denied) ▾

Run Vector Search

\$vectorSearch

— numCandidates: 80, limit: 3 (policies), limit: 3 (prior claims), filter: { plan_type: "PPO", state: "OH", adjudication_outcome: "any" }

WALKTHROUGH

SCREEN 05 / 07

Three policies, three prior cases.

The retrieval returns the coverage policies that govern this scenario and the comparable prior claims — both approved and denied — ranked by semantic similarity to the current note.

POLICIES Top-3 by similarity, auto-scoped to clinical_area

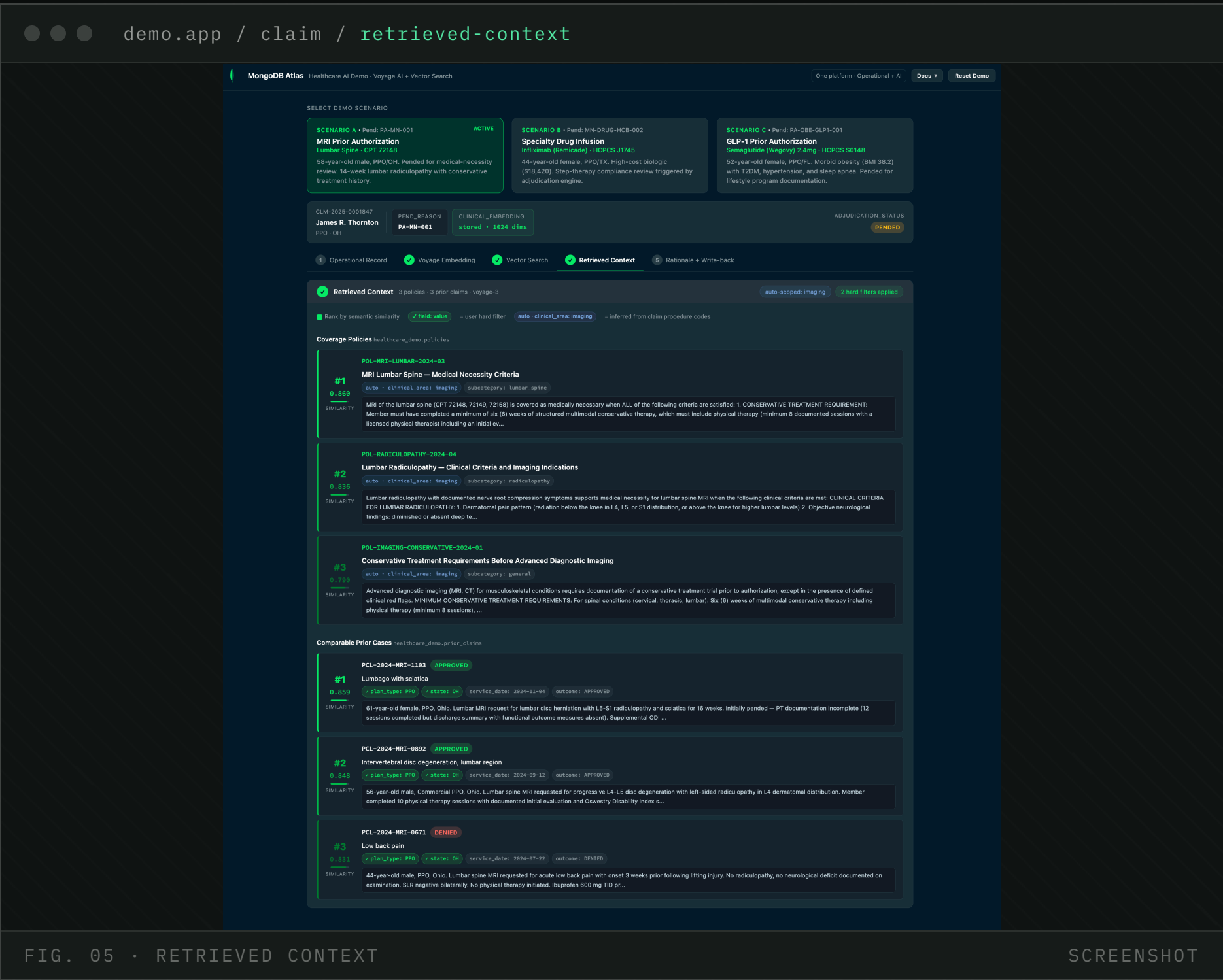
PRIOR CASES Top-3 with explicit outcome labels (approved / denied)

PROVENANCE Every result has a stable ID for write-back

Vector Search

similarity score

document IDs



WALKTHROUGH

SCREEN 06 / 07

Generate a reviewer-voice rationale.

LLM grounded prompt

 **MongoDB Atlas**

Healthcare AI Demo · Voyage AI + Vector Search

One platform · Operational + AI

Docs ▾

Reset Demo

SELECT DEMO SCENARIO

SCENARIO A · Pend: PA-MN-001

ACTIVE

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PPO · OH

PEND_REASON
PA-MN-001

CLINICAL_EMBEDDING
stored · 1024 dims

ADJUDICATION_STATUS
PENDING

1 Operational Record

✓ Voyage Embedding

✓ Vector Search

✓ Retrieved Context

5 Rationale + Write-back

5 AI Rationale + Write-back

Template generates · MongoDB stores · claim updates in place

Generate Rationale

Click **Generate Rationale** to assemble a structured reviewer-voice recommendation grounded in the retrieved policy criteria and prior case analogues, then write it back into the same claim record. Watch the claim document above update in place — adjudication status, rationale, and timestamp all written to the same MongoDB document that held the operational data.

WALKTHROUGH

SCREEN 07 / 07

Write-back — in place, in the same document.

Recommendation, policy IDs, prior-case IDs, and a new status history entry are written back into the same claim record — operational data and AI output in one document.

- STATUS

adjudication_status → READY_FOR_REVIEW
- RATIONALE

Full reviewer-voice text stored at claims.ai_rationale
- PROVENANCE

Supporting policy & prior-case IDs persisted alongside

Atlas

single document update

audit log

demo.app / claim / complete

MongoDB Atlas

Healthcare AI Demo · Voyage AI + Vector Search

One platform · Operational + AI

Docs

Reset Demo

SELECT DEMO SCENARIO

SCENARIO A · Pend: PA-MN-001

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CLM-2025-0001847

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PPO · OH

PEND_REASON

PA-MN-001

CLINICAL_EMBEDDING

stored · 1024 dims

AI_DETERMINATION

ADDITIONAL INFORMATION REQUIRED

ADJUDICATION_STATUS

READY FOR REVIEW

1 Operational Record

Voyage Embedding

Vector Search

Retrieved Context

Rationale + Write-back

AI Rationale + Write-back

Template generates · MongoDB stores · claim updates in place

Regenerate

AI Recommendation: **ADDITIONAL INFORMATION REQUIRED**

Draft · Human reviewer attestation required

SUPPORTING POLICIES WRITTEN TO RECORD

POL-MRI-LUMBAR-2024-03

POL-RADICULOPATHY-2024-04

POL-IMAGING-CONSERVATIVE-2024-01

COMPARABLE CASES WRITTEN TO RECORD

PCL-2024-MRI-1103

PCL-2024-MRI-0892

PCL-2024-MRI-0671

Full rationale (written to claims.ai_rationale):

DRAFT CLINICAL RECOMMENDATION

Claim: CLM-2025-0001847

Member: HUM-OH-4482019-01

Generated: 2026-05-22 00:05:49 UTC

Recommendation: ADDITIONAL INFORMATION REQUIRED

DETERMINATION

Authorization request for MRI Lumbar Spine without contrast is pending receipt of complete physical therapy documentation. Clinical notes document a conservative treatment course consistent with plan criteria under POL-MRI-LUMBAR-2024-03 — physical therapy sessions, NSAID trial, and chiropractic were completed — and objective examination findings (positive straight leg raise, dermatomal sensory deficit) support the clinical indication of lumbar radiculopathy. However, the physical therapy record submitted does not include functional outcome measures (e.g., Oswestry Disability Index or equivalent) at discharge, which are required to confirm completion of an adequate therapy course. Upon receipt of the complete PT documentation package including initial evaluation and functional outcome scores, this case is likely to meet medical necessity criteria and advance to authorization.

SUPPORTING POLICY BASIS

Policy: POL-MRI-LUMBAR-2024-03 — MRI Lumbar Spine — Medical Necessity Criteria

Applicable criterion: CONSERVATIVE TREATMENT REQUIREMENT: Member must have completed a minimum of six (6) weeks of structured multimodal conservative therapy, which must include physical therapy (minimum 8 documented sessions with a licensed physical therapist including an initial evaluation, session notes, and functional outcome measures) AND at least one of: NSAIDs at therapeutic dose, muscle relaxants, or chiropractic manipulation.

COMPARABLE PRIOR CASES

PCL-2024-MRI-1103 (APPROVED, PPO/OH) — 1 lumbar with scoliosis. Initially pending for incomplete PT documentation (missing functional outcome measures).

Write-back complete. The claim record above now reflects: adjudication_status: READY_FOR_REVIEW, the full rationale text, supporting policy IDs, comparable case IDs, and an updated status history entry — all written back to the same MongoDB document that holds the operational data. Operational data, embeddings, retrieval, and AI output: one platform.

FIG. 07 · WRITE-BACK COMPLETE

SCREENSHOT

HEALTHCARE-DEMO · COMPLETE

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Q & A

Thank *you.*

Atlas

Voyage AI

Vector Search